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#include<iostream>
#include<fstream>
#include<iomanip>
using namespace std;

void readFile(fstream& infile, float farenheit[], int *total);
void computeC(float farenheit[], double celcius[], int total);
float average(double celcius[], int total);
void grade(char tempgrade[], double celcius[], int total, int *a, int *b, int *c);
void writeFile(fstream& outfile, double celcius[], float farenheit[], int total, char tempgrade[]);

int main()
{
    fstream infile("input.txt", ios::in);
    fstream outfile("output.txt", ios::out);
    float farenheit[8], avg;
    double celcius[8];
    int total =0, a=0, b=0, c=0;
    char tempgrade[8];
    float trialcelcius[8];

    if(!infile)
    {
        cout<<"error when opening files";
        exit(1);
    }

    readFile(infile, farenheit, &total);

    computeC(farenheit, celcius, total);

    avg = average(celcius, total);

    grade(tempgrade, celcius, total, &a, &b, &c);
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    writeFile(outfile, celcius, farenheit, total, tempgrade);

    infile.close();
    outfile.close();

    cout<<"Average of temperature in celcius: "<<avg<<endl;
    cout<<"Number of high temperature: "<<a<<endl;
    cout<<"Number of medium temperature: "<<b<<endl;
    cout<<"Number of low temperature: "<<c<<endl;

    return 0;
}

void readFile(fstream& infile, float farenheit[], int *total)
{
    for(int i = 0; i<8; i++)
    {
        infile>>farenheit[i];
        ++*total;
    }
}

void computeC(float farenheit[], double celcius[], int total)
{
    for(int i=0; i<total; i++)
    {
        celcius[i] = (farenheit[i] - 32.0)*(5.0/ 9.0);

    }
}

float average(double celcius[], int total)
{
```

```
    float totaltemp=0, average;

    for(int i = 0; i<total; i++)
    {
        totaltemp = totaltemp + celcius[i];
    }
    average = totaltemp/total;
    cout<<fixed<<setprecision(1);

    return average;
}

void grade(char tempgrade[], double celcius[], int total, int *a, int *b, int *c)
{
    for(int i=0; i<total; i++)
    {
        if(celcius[i] >= 35.0)
        {
            tempgrade[i] = 'H';
            ++*a;
        }
        else if(celcius[i] < 35.0 && celcius[i] >= 20.0)
        {
            tempgrade[i] = 'M';
            ++*b;
        }
        else
        {
            tempgrade[i] = 'L';
            ++*c;
        }
    }
}

```

```
void writeFile(fstream& outfile, double celcius[], float farenheit[], int total, char tempgrade[])
{
    outfile<<"C(Celcius)\t\tF(Farenheit)\t\tDescription"<<endl;
    outfile<<"==========\t\t============\t\t============"<<endl;
    for(int i=0; i<total; i++)
    {
        outfile<<fixed<<setprecision(2)<<celcius[i]<<"\t\t\t"<<farenheit[i]<<"\t\t\t"<<tempgrade[i]<<endl;
    }
}
```